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Introduction

The HP 848x series sensors are prized in amateur circle. Unfortunately they are easily damaged by too much RF. Whilst it's risky to dismantle a working sensor, curiosity makes us wonder about what's inside.

This is a summary of the steps involved in opening a HP8482A sensor. It shows things to look out for during the dismantling process and of course, what it looks like inside.

Further reading

<https://literature.cdn.keysight.com/litweb/pdf/08481-90173.pdf?id=438053>

1. Dismantle the case



The case is a two part plastic assembly with clips running along each side. Push a flat bladed screwdriver into the gaps of the case at both ends of the sensor. It will require some force to get the shell apart. There is no need to remove the labels on each side of the case.

Two metal shims wrap around the body of the sensor. Mind your fingers, they're sharp.

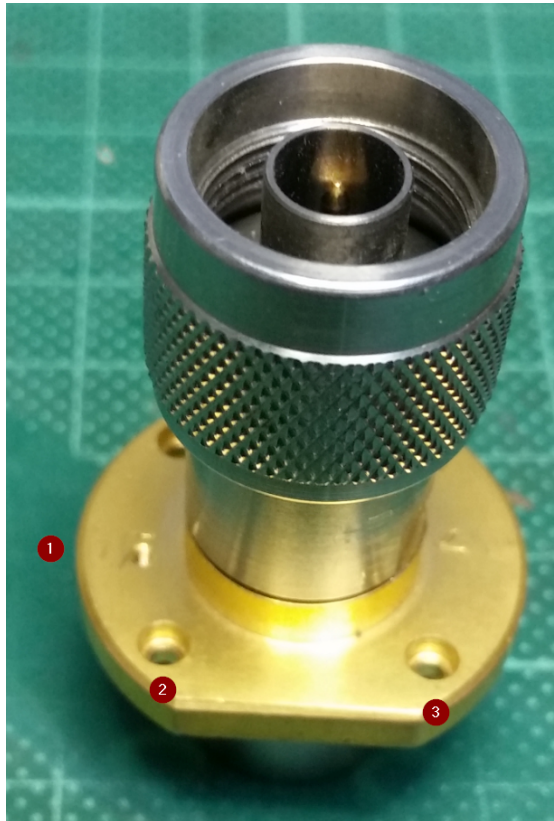
Disclaimer

Don't try this with a working sensor. It is a precision instrument and a moment of inattention could render it useless.

2. Dismantling the sensor body.

Take care in this step as you can permanently damage the sensor if you're not careful.

The N connector assembly (including thermistor), the PCB and the 12 pin cable connector are all screwed together to the bottom shell of the sensor body. You must identify the bottom shell before you remove any screws.



In the image at left, hole number one indicates where the N connector assembly is screwed to the bottom shell of the sensor. The screw is fitted from the inside of the sensor.

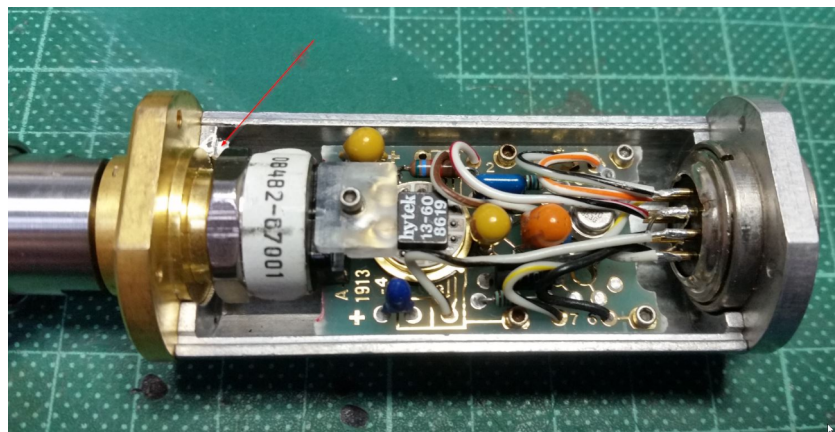
Holes 2 & 3 show where the two mounting screws hold the N connector assembly to the bottom shell of the sensor.

At the rear of the sensor you should now be able to identify the screws that hold the back plate including 12 pin connector, to the bottom shell.

DO NOT remove the screws for the bottom shell. This might cause permanent damage to the sensor.

Just remove the four screws holding the top shell of the sensor with a 0.050" Allen key then lever the shell off carefully.

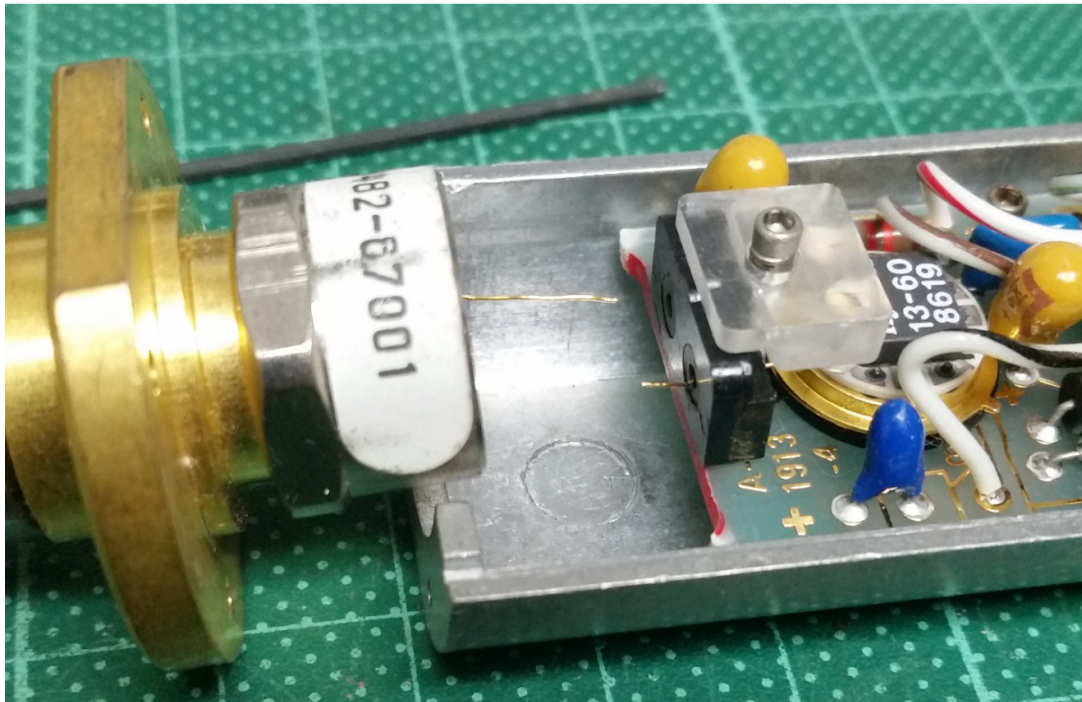
3. Inside the sensor body.



Shown above is the inside of the sensor. The arrow indicates the location of the screw that holds the N connector assembly to the bottom shell.

HP advises that the wires attached to the PCB should not be moved as it may affect calibration.

4. Thermistor mount connections



Above shows the N connector assembly detached from the sensor body. The fine wires pass through two small ferrite beads and are then held against the chopper by the clear plastic clamp. It should be clear now how easily it is to damage the probe if the screws for the bottom shell are removed.

5. Exploded view of the clamp assembly.



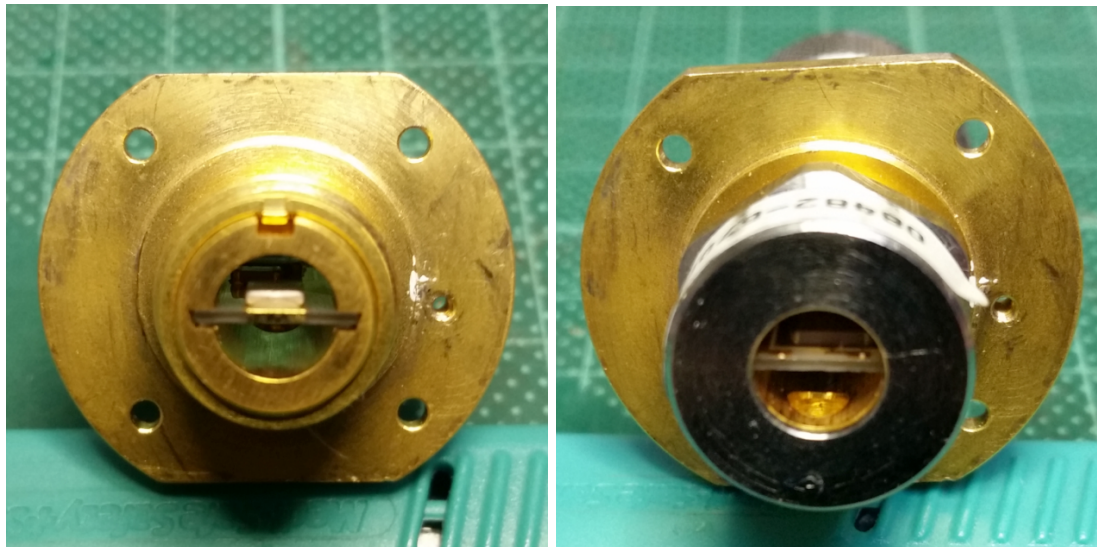
A view of the clamp components and the chopper device. The wires from the thermistor clamp onto the two gold pads on the chopper.

6. Exploded view of the N connector assembly.



It seems this part of the sensor is tailor made for the specific model. The locking ring at right has a label indicating it is a HP8482A sensor. The thermistor mount has a small PCB inside (looks like glass) holding a number of components. At the left of the assembly there is a small gold leaf that presses against the spring contact of the N connector pin.

7. Thermistor



Best focus I could manage, here we see the connections to the thermistor. Four tiny screws hold the board in the mount.

Summary

This sensor was definitely damaged before I started. Whilst it is possible to dismantle a sensor safely there is always a risk that something can go wrong rendering it useless.

Dropping the sensor could bend or break the fine gold wires from the thermistor. It's not clear if there is any precision required in clamping the mounting wires or if the sensor characteristic would change if they were moved.